EXPERIMENT 17

Lexical Database Access and Synset Exploration using WordNet

AIM

To query the WordNet lexical database using NLTK in Python to retrieve synsets, definitions, examples, hypernyms, and calculate semantic similarity.

PROCEDURE

1. Import `nltk.corpus.wordnet as wn`.
2. Query synsets for a target polysemous lemma.
3. Inspect definitions, lemmas, and usage examples for each synset.
4. Traverse the hypernym (is-a) taxonomic hierarchy.
5. Calculate path-based semantic similarity between noun pairs.

PROGRAM

```
import nltk
from nltk.corpus import wordnet as wn

nltk.download('wordnet', quiet=True)

query_word = "bank"
synsets = wn.synsets(query_word)

print(f"=== Synsets for '{query_word}' (Total found: {len(synsets)}) ===")
for i, syn in enumerate(synsets[:4], 1):
    print(f"\n{i}. Synset: {syn.name()} ({syn.pos()})")
    print(f"   Definition: {syn.definition()}")
    print(f"   Lemmas: {[l.name() for l in syn.lemmas()]}")
    if syn.examples():
        print(f"   Example: \"{syn.examples()[0]}\"")

tree_synset = wn.synset('bank.n.01')
print(f"\nHypernyms for {tree_synset.name()}:")
for parent in tree_synset.hypernyms():
    print(f" -> {parent.name()}: {parent.definition()}")

w1 = wn.synset('cat.n.01')
w2 = wn.synset('dog.n.01')
w3 = wn.synset('computer.n.01')

print("\n=== Semantic Similarity Scores ===")
print(f"Similarity (cat, dog): {w1.path_similarity(w2):.4f}")
print(f"Similarity (cat, computer): {w1.path_similarity(w3):.4f}")
```

OUTPUT

```
> python3 experiment_17.py
--------------------------------------------------------------------------------
=== Synsets for 'bank' (Total found: 18) ===

1. Synset: bank.n.01 (n)
   Definition: sloping land (especially the slope beside a body of water)
   Lemmas: ['bank']
   Example: "they pulled the canoe up on the bank"

2. Synset: depository_financial_institution.n.01 (n)
   Definition: a financial institution that accepts deposits and channels the money into
lending activities
   Lemmas: ['depository_financial_institution', 'bank', 'banking_concern',
'banking_company']
   Example: "he cashed a check at the bank"

3. Synset: bank.n.03 (n)
   Definition: a long ridge or pile
   Lemmas: ['bank']
   Example: "a bank of earth"

4. Synset: bank.n.04 (n)
   Definition: an arrangement of similar objects in rows or tiers
   Lemmas: ['bank']
   Example: "he operated a bank of switches"

Hypernyms for bank.n.01:
 -> slope.n.01: an elevated geological formation

=== Semantic Similarity Scores ===
Similarity (cat, dog): 0.2000
Similarity (cat, computer): 0.0556
[Process completed with exit code 0]
```

RESULT

Thus, lexical queries, synset exploration, ontological traversal, and semantic similarity scoring were successfully performed using WordNet.